

L9b: Black–Scholes–Merton Premiums and Pricing American Style Options Contracts using the CRR Binomial Model

CHEME 5660 Cornell University

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Objectives for Today

Previously, we priced European options, which can be exercised only at expiration. However, standard options on stocks and ETFs are American-style and can be exercised at any time. What is the right to exercise early worth?

Today's concepts

- Price European options with the Black–Scholes–Merton model: closed-form call and put premiums and implied volatility.
- Apply the binomial model to price American options: build CRR trees and compute premiums with the decision rule for early exercise.
- Estimate the premium of American options by simulation: Longstaff–Schwartz Monte Carlo, regressing the value of holding on the share price.

Lectures and Examples

Lecture:
CHEME-5660-L9b-Lecture-American-Derivatives-Intrinsic-Extrinsic-Value-Fall-2026.ipynb

Concept Review example:
CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

Example 1:
CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Example 2:
CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Review on your own:
CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

Concept Review: The European Premium

How do we compute the premium of a European option? Previously, we set the contract's expected NPV to zero under the risk-neutral pricing rule $\mathbb{Q}$, under which the expected share price grows at the risk-free rate g_y. For a call or a put with payoff $V_\star(K, S(T))$:

$$\boxed{\mathcal{P}_\star^{\mathsf{EU}} = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}\big[V_\star(K, S(T))\big] \quad \text{where } \star \in \{c, p\}}$$

where K is the strike price, $S(T)$ is the share price at expiration T (years), and $\mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y T}$ discounts the payoff back to today.

The European premium is the discounted expected payoff under $\mathbb{Q}$, one price for all buyers. Monte Carlo simulation and the Black–Scholes–Merton formulas are two ways to compute it.

Concept Review: A Monte Carlo Premium

Simulation estimates the market premium directly. The payoff depends only on $S(T)$, so we draw it from the solution of geometric Brownian motion (GBM) with drift g_y and volatility σ.

Initialize: given $S(0)$, K, the payoff $V(K, S)$, T (years), σ, g_y, and M paths.

1. For each path $i = 1, 2, \ldots, M$ **do**:
   a. Draw a standard normal random number Z_i.
   b. $S_i(T) \leftarrow S(0) \cdot \exp\left[(g_y - \sigma^2/2)\,T + \sigma\sqrt{T}\,Z_i\right]$.
   c. $V_i \leftarrow V(K, S_i(T))$.
2. Discount the average payoff: $\mathcal{P} \leftarrow \mathcal{D}_{T,0}^{-1}(g_y) \cdot \frac{1}{M}\sum_{i=1}^{M} V_i$.

The result is an estimate: it changes with the random draws, and it becomes more precise as M grows.

Example: CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

Optional: an alternative way to think about the premium, in the contingent claims notebook.

Company Profile: Options Clearing Corporation

OCC stands between the buyer and the seller of every U.S. listed option. It opened in 1973 as CBOE Clearing Corporation and became OCC in 1975.

- Exercise and assignment: OCC assigns each early exercise at random to a clearing member with a short position in the same contract.
- Automatic exercise at expiration: since 1977, an expiring equity option at least 0.01 USD/share in the money is exercised unless the holder instructs otherwise.
- Monte Carlo margin and binomial trees: since 2006, STANS simulates 10,000 two-day scenarios and reprices American options on a binomial tree.

Connection to today's lecture: a seller of an American put can be assigned once exercising now beats holding, so a European-premium margin would leave out the early-exercise premium.

The Black–Scholes–Merton Equation

Merton and Scholes received the 1997 Nobel Prize in Economic Sciences for this model. Fischer Black had passed away before the award.

Assume the price of a non-dividend-paying share follows GBM with constant σ and a constant, continuously compounded risk-free rate g_y. The premium $\mathcal{P}(S,t)$, with $\mathcal{P}(S,T) = V(K,S)$, satisfies:

$$\boxed{\frac{\partial \mathcal{P}}{\partial t} + g_y S \frac{\partial \mathcal{P}}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \mathcal{P}}{\partial S^2} = g_y \mathcal{P}}$$

Solving it backward in time from the payoff gives today's premium, $\mathcal{P}(S(0),0)$, and this solution equals the risk-neutral premium $\mathcal{P}^{\text{EU}}$ that the Monte Carlo simulation estimates.

Derivation: the contingent claims notebook.

Black–Scholes–Merton Premiums

Instead of simulating, we can compute this premium with the closed-form European call and put formulas:

$$\boxed{\begin{aligned}\mathcal{P}_c(K, S(0)) &= N(d_+) \cdot S(0) - N(d_-) \cdot K \cdot \mathcal{D}_{T,0}^{-1}(g_y) \\ \mathcal{P}_p(K, S(0)) &= N(-d_-) \cdot K \cdot \mathcal{D}_{T,0}^{-1}(g_y) - N(-d_+) \cdot S(0)\end{aligned}}$$

where N is the standard normal CDF, T is in years, and:

$$d_+ = \frac{1}{\sigma\sqrt{T}}\left[\ln\left(\frac{S(0)}{K}\right) + \left(g_y + \frac{\sigma^2}{2}\right)T\right], \qquad d_- = d_+ - \sigma\sqrt{T}$$

We know $S(0)$, K, T, and g_y when we price the contract. The volatility σ is the one input we must choose.

Example: CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

Which Volatility Should We Use?

If we knew the right volatility, the BSM model (or the Monte Carlo model) would compute the observed market premium. In practice, we observe the market premium and back out the volatility that makes the model match it.

- Historical volatility: $\sigma_{\text{ann}} = \sqrt{\Delta t} \cdot \sigma_g$, where σ_g is the standard deviation of past growth rates (units: yr^{-1}) and Δt is the time between observations (units: years). It looks backward.
- Implied volatility (IV): the market's forecast of a likely future movement in the share price. It looks forward, and it is often at or above the historical volatility.
- How can we use it? IV represents a one-standard-deviation movement in the share price over the next year. Over T years, the move scales with $\sqrt{T}$.

This $\sqrt{T}$ scaling gives a quick range for the share price on the next slide.

A Back-of-the-Envelope Range from IV

Moving one standard deviation up or down in the solution of geometric Brownian motion, $Z = \pm 1$, gives a back-of-the-envelope range for the share price:

$$
\begin{aligned}
S(T) &= S(0)\exp\left[\mu_g T + \sigma\sqrt{T}\,Z\right] && \text{solution of geometric Brownian motion} \\
S_{\pm} &= S(0)\exp\left[\mu_g T \pm \mathsf{IV}\sqrt{T}\right] && \text{set } \sigma = \mathsf{IV} \text{ and } Z = \pm 1 \\
&\approx S(0)\exp\left[\pm \mathsf{IV}\sqrt{T}\right] && \mu_g T \text{ is small next to } \mathsf{IV}\sqrt{T} \text{ for short } T \\
&\approx S(0)\left(1 \pm \mathsf{IV}\sqrt{T}\right) && e^x \approx 1 + x \text{ for small } x
\end{aligned}
$$

where μ_g is the mean growth rate, Z is standard normal, and $S_{\pm}$ is the share price one standard deviation above or below $S(0)$. Under the model, the share price ends inside this range about 68% of the time.

Implied Volatility by Bisection

We back out the implied volatility $\hat{\sigma}$ by bisection.

Initialize: given $\mathcal{P}^{\text{market}}$, $S(0)$, K, T (years), g_y, a bracket $\sigma_{\text{lo}} < \sigma_{\text{hi}}$ with $\mathcal{P}(\sigma_{\text{lo}}) \leq \mathcal{P}^{\text{market}} \leq \mathcal{P}(\sigma_{\text{hi}})$, a tolerance $\epsilon > 0$, `maxiter`, and $i \leftarrow 1$.

While $\sigma_{\text{hi}} - \sigma_{\text{lo}} > \epsilon$ and $i \leq$ `maxiter` **do**:

1. $\sigma_{\text{mid}} \leftarrow (\sigma_{\text{lo}} + \sigma_{\text{hi}})/2$ and compute its BSM premium $\mathcal{P}(\sigma_{\text{mid}})$.
2. If $\mathcal{P}(\sigma_{\text{mid}}) < \mathcal{P}^{\text{market}}$, **then** $\sigma_{\text{lo}} \leftarrow \sigma_{\text{mid}}$. Otherwise, $\sigma_{\text{hi}} \leftarrow \sigma_{\text{mid}}$.
3. $i \leftarrow i + 1$.

The implied volatility is the midpoint of the final bracket, $\hat{\sigma} \leftarrow (\sigma_{\text{lo}} + \sigma_{\text{hi}})/2$. The BSM premium rises with σ, so exactly one σ matches the market premium, and each step halves the bracket around it.

Implied Volatility in Practice

Three points about using implied volatility with real quotes:

- Market quotes: the SPXW volatility skew notebook applies this bisection to a chain of S&P 500 index options.
- One per contract: each strike and expiration has its own implied volatility, although the BSM model assumes one constant σ. This pattern is known as the volatility smile or volatility skew.
- Pricing without a quote: to price a contract without a quote, or on a future date, we can predict its implied volatility from a surface fitted to today's quotes, which gives implied volatility as a function of strike and expiration (Sun et al., 2026).

Next, we link the call and put premiums with the same strike and expiration.

Optional: the SPXW volatility skew notebook.

Interesting Fact: Put–Call Parity

For a European call and put with the same strike and expiration, the payoffs differ by:

$$V_c(K, S(T)) - V_p(K, S(T)) = \max(S(T) - K,\ 0) - \max(K - S(T),\ 0) = S(T) - K$$

$$\begin{aligned}
\mathcal{P}_c(K, S(0)) - \mathcal{P}_p(K, S(0)) &= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_c(K, S(T))] && \text{premium formula} \\
&\quad - \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_p(K, S(T))] \\
&= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_c(K, S(T)) - V_p(K, S(T))] && \text{expectation is linear} \\
&= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[S(T) - K] && \text{payoff identity} \\
&= \mathcal{D}_{T,0}^{-1}(g_y) \cdot (S(0)\, e^{g_y T} - K) && \text{grows at } g_y \text{ under } \mathbb{Q} \\
&= S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y) && \mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y T}
\end{aligned}$$

This is put–call parity: $\boxed{\mathcal{P}_c(K, S(0)) - \mathcal{P}_p(K, S(0)) = S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y)}$

It holds only for European contracts: an American holder can exercise early.

Interesting fact: CHEME-5660-L9b-InterestingFact-PutCallParity-Fall-2026.ipynb

American Style Options Contracts

American-style contracts can be exercised at any time before or on the expiration date T. We write the American premium as:

$$\mathcal{P}_\star^{\text{AM}} = \max_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\Big(\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V_\star(K, S(\tau))\Big) \geq \mathcal{P}_\star^{\text{EU}} \qquad \text{for } \star \in \{c, p\}$$

- Stopping time: τ uses only the information available at that time.
- Discounting: $\mathcal{D}_{\tau,0}^{-1}(g_y)$ brings the payoff at strike price K back to today.
- At least the European premium: the holder can always wait.

The standard pricing method is the Cox, Ross, Rubinstein (CRR) binomial model.

Reference: Cox, Ross, and Rubinstein (1979). Option pricing: A simplified approach. J. Financial Economics 7(3): 229.

CRR Binomial Lattice Model

The share price moves up by u with risk-neutral probability q, or down by d:

$$
\begin{aligned}
\mathcal{D}_{1,0}(g_y) \cdot S_0 &= \mathbb{E}_{\mathbb{Q}}\left(S_1\right) && \text{grow at the risk-free rate } g_y \text{ under } \mathbb{Q} \\
&= q \cdot (u \cdot S_0) + (1-q) \cdot (d \cdot S_0) && \text{up with probability } q\text{, down with } (1-q) \\
&= S_0(qu + (1-q)d) && \text{factor out } S_0 \\
&= S_0(qu + d - qd) && \text{expand } (1-q)d \\
&= S_0(d + q(u-d)) && \text{collect the } q \text{ terms}
\end{aligned}
$$

$$
\implies \boxed{q = \frac{\mathcal{D}_{1,0}(g_y) - d}{u - d}} \qquad \text{divide by } S_0 \text{ and solve for } q
$$

where S_0 is today's share price and $R_f = \mathcal{D}_{1,0}(g_y) = e^{g_y \Delta t}$ for one step Δt.

CRR Up and Down Factors

With $ud = 1$, let $a = \ln u = -\ln d$. Match the growth rate $g = \pm a/\Delta t$ to GBM:

$$\frac{(2p-1)\,a}{\Delta t} = \mu_g \qquad \text{match the mean growth rate}$$

$$\frac{4p(1-p)\,a^2}{\Delta t^2} = \frac{\sigma^2}{\Delta t} \qquad \text{match the variance}$$

$$\frac{a^2}{\Delta t^2} - \mu_g^2 = \frac{\sigma^2}{\Delta t} \qquad \text{use } 4p(1-p) = 1-(2p-1)^2 \text{ and the mean}$$

$$a^2 = \sigma^2 \Delta t + \mu_g^2 \Delta t^2 \qquad \text{solve for } a^2$$

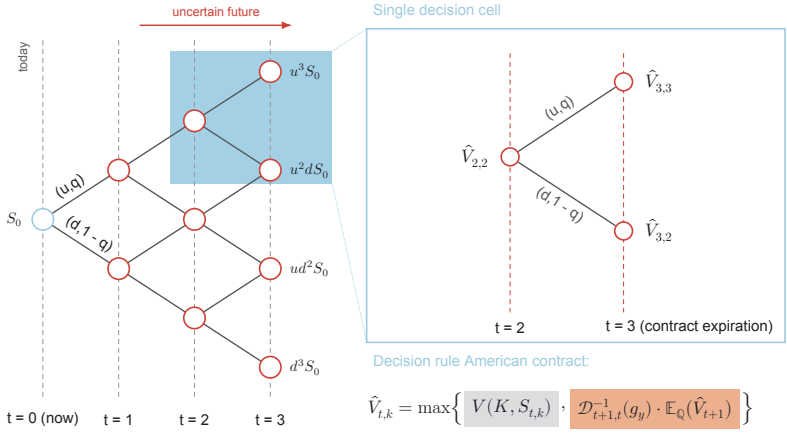
$$\implies \boxed{u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}} \qquad \text{keep the first term, so } a = \sigma\sqrt{\Delta t}$$

where p is the real-world up probability, μ_g the mean growth rate, and σ the volatility.

Where do u and d come from? advanced/crr_factors/CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

Decision-Rule for American Options

At each node, the American holder decides to exercise now or hold for the future.



The Early-Exercise Premium

Working backward from expiration, the American holder takes the larger of exercising and holding, while the European holder can only hold:

$$\hat{V}^{\mathsf{AM}}_{t,k} = \max\Big\{\underbrace{V(K, S_{t,k})}_{\text{exercise now}},\ \underbrace{\mathcal{D}^{-1}_{t+1,t}(g_y)\cdot\overbrace{\left(q\,\hat{V}^{\mathsf{AM}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{AM}}_{t+1,k}\right)}^{\mathbb{E}_{\mathbb{Q}}}}_{\text{hold for the future}}\Big\}$$

$$\hat{V}^{\mathsf{EU}}_{t,k} = \mathcal{D}^{-1}_{t+1,t}(g_y)\cdot\overbrace{\left(q\,\hat{V}^{\mathsf{EU}}_{t+1,k+1} + (1-q)\,\hat{V}^{\mathsf{EU}}_{t+1,k}\right)}^{\mathbb{E}_{\mathbb{Q}}}$$

where $\hat{V}_{t,k}$ is the contract's value with k up moves in t steps, $S_{t,k} = S_0 u^k d^{t-k}$, and $\mathcal{D}^{-1}_{t+1,t}(g_y) = e^{-g_y \Delta t}$. The premium is $\mathcal{P} = \hat{V}_{0,0}$.

Early-exercise premium: $\mathcal{P}^{\mathsf{AM}} - \mathcal{P}^{\mathsf{EU}} \geq 0$.

Pseudocode: Pricing on the CRR Lattice

Initialize: given S_0, K, the payoff function $V(K, S)$ of the call or put, T, σ, g_y, and the number of time steps h. Compute $\Delta t = T/h$, u and d, q, and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

1. Build the share price lattice: $S_{t,k} \leftarrow S_0 u^k d^{t-k}$ for each level $t = 0, 1, \ldots, h$ and node $k = 0, 1, \ldots, t$.
2. At expiration, set each node value to the payoff: $\hat{V}_{h,k} \leftarrow V(K, S_{h,k})$ for $k = 0, 1, \ldots, h$.
3. For each level $t = h-1, h-2, \ldots, 0$ **do**:
   - For each node $k = 0, 1, \ldots, t$ **do**:
     a. Compute the value of holding for the future: $H_{t,k} \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot (q\,\hat{V}_{t+1,k+1} + (1-q)\,\hat{V}_{t+1,k})$.
     b. If the contract is American, **then**: $\hat{V}_{t,k} \leftarrow \max\{V(K, S_{t,k}),\ H_{t,k}\}$. Otherwise, $\hat{V}_{t,k} \leftarrow H_{t,k}$.
4. The premium is the value at the root: $\mathcal{P} \leftarrow \hat{V}_{0,0}$.

A Hypothetical Example from Hull

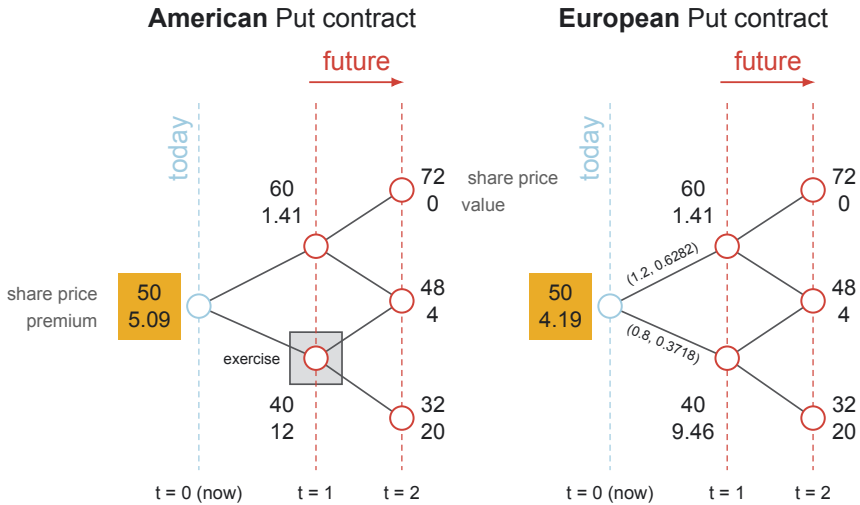
Consider a put contract.

- Contract: the strike price is $K = 52$ USD/share on a stock whose current price is $S_0 = 50$ USD/share.
- Lattice: two time steps of 1 year each. In each period, the stock price moves up by a factor of $u = 1.2$ or down by a factor of $d = 0.8$.
- Pricing inputs: the continuously compounded risk-free rate is $g_y = 5\%$, so the risk-neutral probability of an up move is $q = (e^{0.05} - 0.8)/(1.2 - 0.8) = 0.6282$.

What is the price of the American and European put option contracts?

Source: Chapter 13 of Hull, Options, Futures, and Other Derivatives, 9th edition (2015).

American versus European Put on One Lattice



Early Exercise in the Hull Example

At the down node at $t = 1$, where $S_{1,0} = 40$ (all values in USD/share):

$$\underbrace{V(K, S_{1,0}) = 52 - 40 = 12}_{\text{exercise now}} \; > \; \underbrace{\mathcal{D}_{2,1}^{-1}(g_y)(q \cdot 4 + (1-q) \cdot 20) = 9.46}_{\text{hold for the future}}$$

so the American holder exercises. That one decision raises the premium:

$$\boxed{\mathcal{P}^{\text{AM}} - \mathcal{P}^{\text{EU}} = 5.09 - 4.19 \approx 0.90}$$

Let's compute every node of this lattice, then apply the same method to contracts from the course's options archive.

Review on your own: CHEME-5660-L9b-Example-Hull-American-Put-Fall-2026.ipynb

We'll work through in lecture: CHEME-5660-L9b-Example-AmericanContractsPremium-Fall-2026.ipynb

Monte Carlo Pricing of American Options

We simulate share price paths under $\mathbb{Q}$, then estimate the value of holding at each date by regressing the discounted cash flows from holding on the current share price across paths. Like the lattice, Longstaff and Schwartz work backward from expiration:

Initialize: given S_0, K, the payoff function $V(K, S)$ of the call or put, T, σ, g_y, the number of exercise dates h after today, and the number of paths M. Compute $\Delta t = T/h$ and $\mathcal{D}_{t+1,t}^{-1}(g_y) = e^{-g_y \Delta t}$.

Reference: Longstaff, F. A., and Schwartz, E. S. (2001). Valuing American options by simulation: a simple least-squares approach. Review of Financial Studies 14(1): 113–147.

Pseudocode: Longstaff–Schwartz

1. Simulate: generate M share price paths under $\mathbb{Q}$ from S_0 on the dates $t = 0, 1, \ldots, h$ (expiration), where $S_i(t)$ is the share price on path i at date t.
2. At expiration, set each path's cash flow to its payoff: $Y_i(h) \leftarrow V(K, S_i(h))$ for $i = 1, 2, \ldots, M$, where $Y_i(t)$ is path i's cash flow valued at date t.
3. For each date $t = h-1, h-2, \ldots, 1$ **do**:
   a. Discount each cash flow back one date: $Y_i(t) \leftarrow \mathcal{D}_{t+1,t}^{-1}(g_y) \cdot Y_i(t+1)$.
   b. Regress: on the in-the-money paths, those with $V(K, S_i(t)) > 0$, fit the value of holding $\hat{C}_t(S) = \beta_0 + \beta_1 S + \beta_2 S^2$ to the pairs $(S_i(t), Y_i(t))$ by least squares.
   c. Decide: for each in-the-money path, if $V(K, S_i(t)) > \hat{C}_t(S_i(t))$, **then**: exercise, so $Y_i(t) \leftarrow V(K, S_i(t))$.
4. Discount the cash flows to today and average them: $\bar{Y} \leftarrow \mathcal{D}_{1,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} Y_i(1)$.
5. The premium is $\mathcal{P}^{\text{AM}} \leftarrow \max\{V(K, S_0),\ \bar{Y}\}$.

The Early-Exercise Premium by Simulation

Let's price an American put by simulation and compare it with the lattice and the European premium. We apply two rules to the same simulated paths:

Rule at each date	Estimates
Longstaff–Schwartz: exercise when $V(K, S_i(t)) > \widehat{C}_t(S_i(t))$	$\mathcal{P}^{\text{AM}}$
Hold on every path	$\mathcal{P}^{\text{EU}}$

$$\text{early-exercise premium} \approx \mathcal{P}^{\text{AM}} - \mathcal{P}^{\text{EU}}$$

Example 2: CHEME-5660-L9b-Example-LSM-Premium-Fall-2026.ipynb

Optional Advanced Material

Three optional notebooks extend this lecture:

Contingent claims: advanced/contingent_claims/
CHEME-5660-L9b-Advanced-ContingentClaims-BSM-Fall-2026.ipynb

We derive where the BSM premiums come from: the risk-neutral probability on a one-step lattice, state prices for event contracts, the BSM equation and its solution, and the call built from a digital claim and an asset-or-nothing claim. We also compare a client's certainty equivalent for a one-dollar bet with its market price.

Volatility skew: advanced/spxw_volatility_skew/
CHEME-5660-L9b-Advanced-SPXW-Volatility-Skew-Fall-2026.ipynb

We apply the implied-volatility bisection to an end-of-day chain of SPXW options and compare the skew with the realized volatility of SPY, which tracks the index.

Optional Advanced Material

CRR factors: advanced/crr_factors/
CHEME-5660-L9b-Advanced-CRR-Factors-Fall-2026.ipynb

In this notebook, we derive the up and down factors of the CRR model by matching the mean and variance of the lattice's one-step growth rate to those of geometric Brownian motion. We then show that the lattice still matches geometric Brownian motion when we price with the risk-neutral probability, and find how small the time step must be to keep that probability between zero and one.

Notebooks: lectures/week-9/L9b/advanced, also linked from the lecture notebook.

Summary

Today we priced European options with the BSM formulas, and American options with the CRR binomial model and Monte Carlo simulation.

Key takeaways

- The Black–Scholes–Merton formulas price European options: they match the Monte Carlo premium, and run backward from a market premium they give the implied volatility.
- Binomial trees price American options: backward induction compares exercising with holding at each node. In Hull's put, one early exercise accounts for the whole American–European gap.
- Simulation estimates the same premium: Longstaff–Schwartz works backward along simulated paths, regressing the value of holding on the share price. Holding on every path gives the European premium.

The early-exercise premium is what the right to exercise before expiration is worth, and the lattice and simulation give two ways to compute it.